

ServiceNow – Power BI Integration

servicenow®

REST API



Steps

1. Navigate to: System Web Services -> REST API Explorer
2. Provide API names as Table API
3. Determine the required table (e.g., /api/now/table/incident)
4. In this case, we are going to use server table (cmdb_ci_server)

REST API Explorer

Namespace
now

API Name
Table API

API Version
latest

► [Retrieve records from a table \(GET\)](#)

Create a record (POST)

Retrieve a record (GET)

Modify a record (PUT)

Delete a record (DELETE)

Update a record (PATCH)

Table API

Allows you to perform create, read, update and delete (CRUD) operations on existing tables

Retrieve records from a table

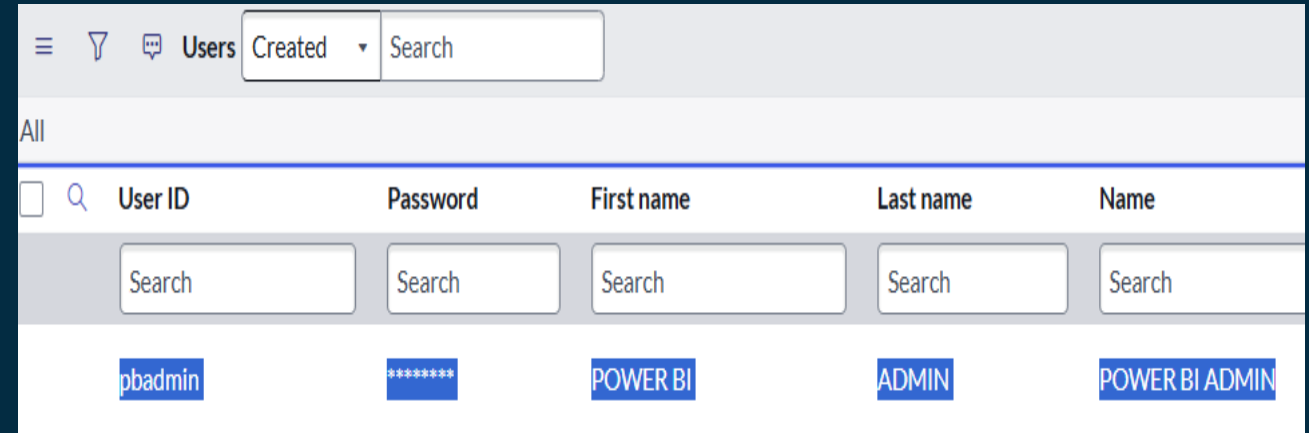
GET `https://dev217783.service-now.com/api/now/table/{tableName}`

Prepare request

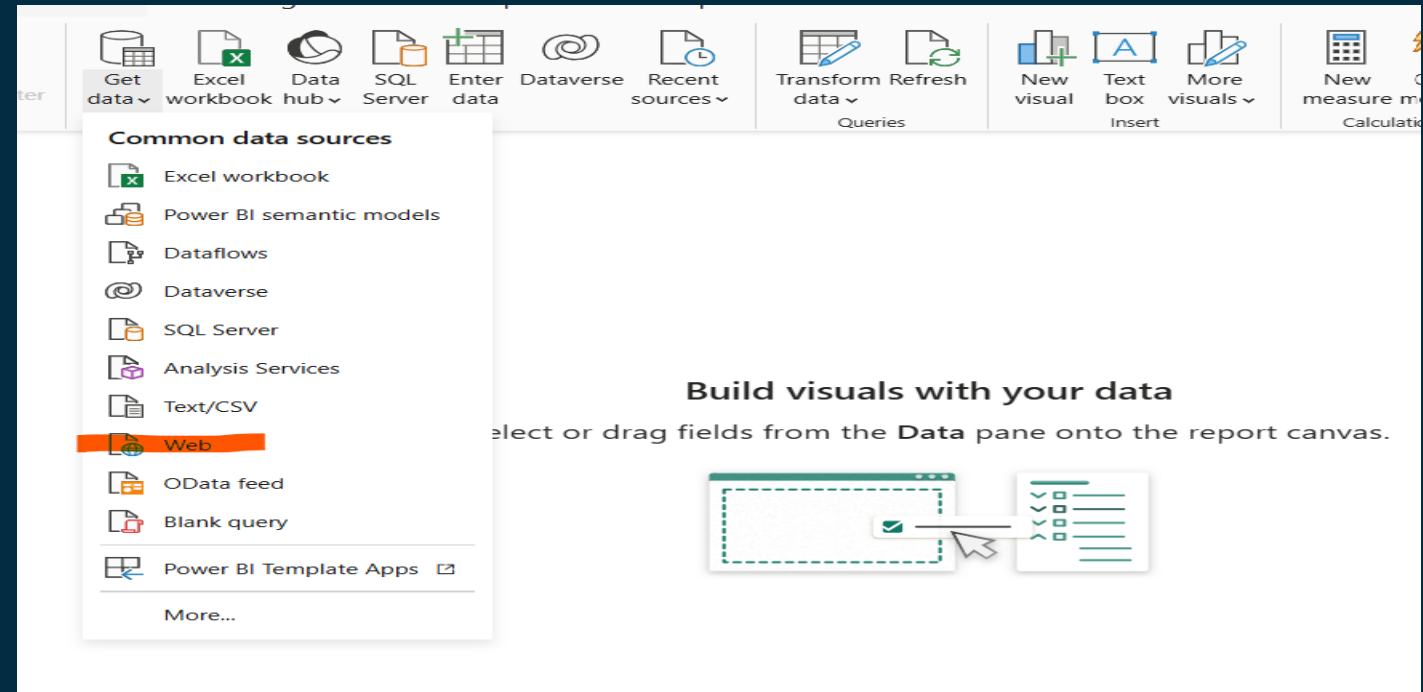
Path parameters

Name	Value
* tableName	Server (cmdb_ci_server)

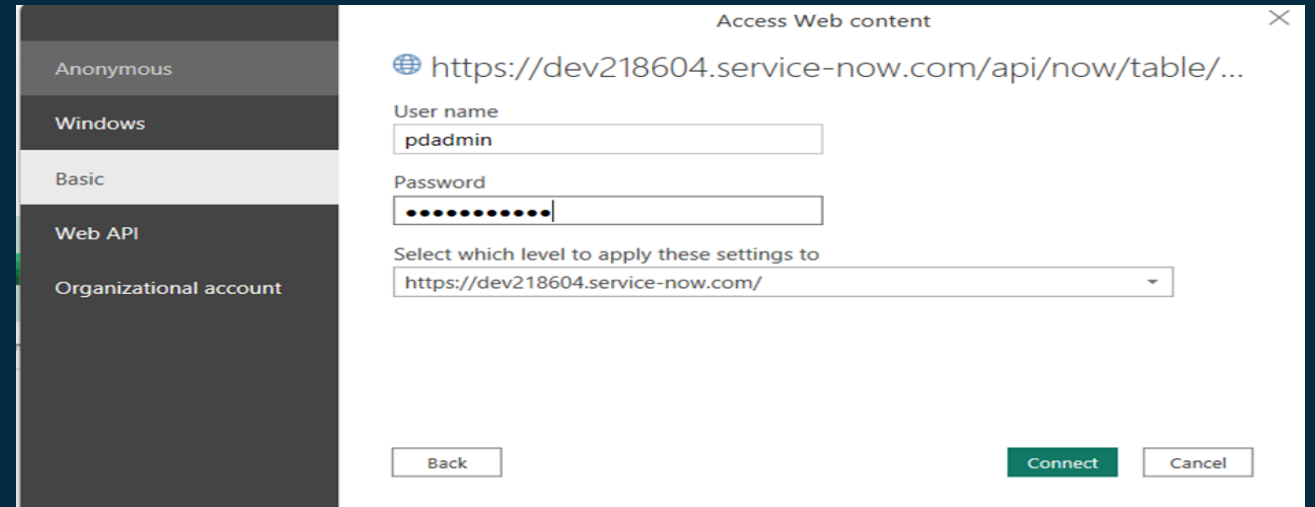
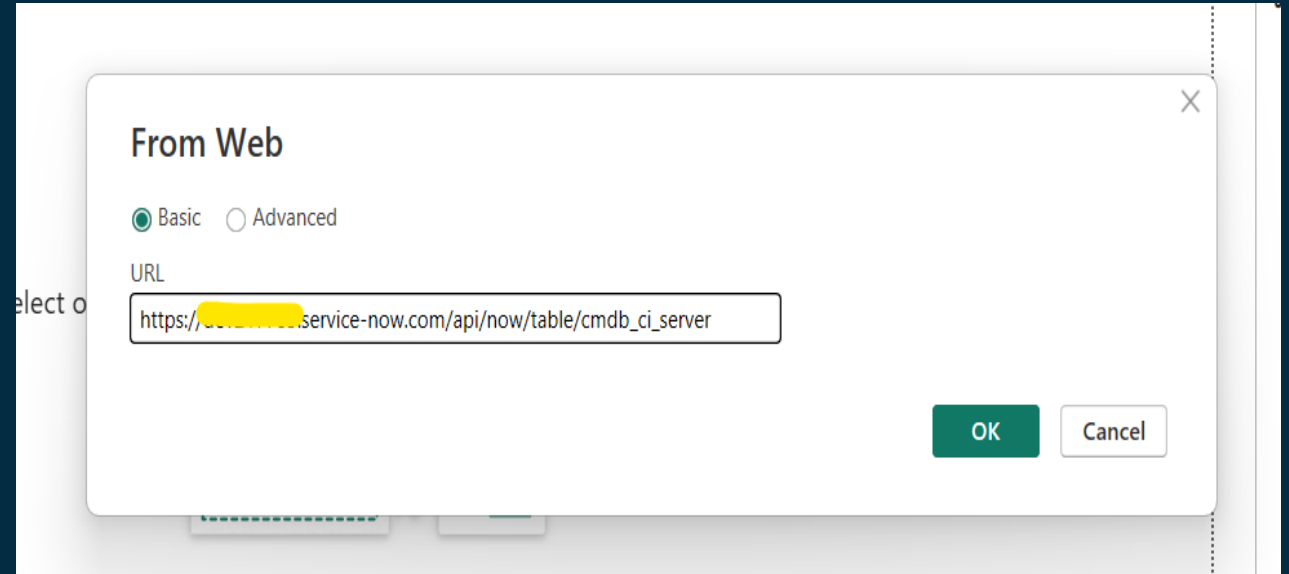
5. Create a user in sys_user table and add the following roles: rest_api_explorer, rest_service
6. Connect to Power BI to ServiceNow
7. Open Power BI Desktop
8. Click on Get Data -> Web



User ID	Password	First name	Last name	Name
pbadmin	*****	POWER BI	ADMIN	POWER BI ADMIN



9. Provide the ServiceNow API
10. Provide the credentials, user which is created in the Step-5
11. Give Basic and fill the User name and Password
12. Click on Connect



13. Once you will click data will be retrieved in Power BI Desktop on right side of the pane in Power BI Workspace

14. Add the filters & slicers as per your requirement

ServiceNow - Power BI Integration

result.name	result.os	result.os_version	result.sys_class_name	result.asset_tag
ApplicationServerHelpdesk	Windows XP		cmdb_ci_server	P1000027
ApplicationServerPeopleSoft	Linux Red Hat		cmdb_ci_server	P1000204
AS400	OS/400	V5R2	cmdb_ci_server	P1000034
Car-1	Windows XP	5.2 build 3790	cmdb_ci_server	P1000171
Car-2	HP/UX	2.6.050	cmdb_ci_server	P1000071
Car-3	Windows XP	5.2 build 3790	cmdb_ci_server	P1000113
Car-4	Windows 2000	5.0 build 2195	cmdb_ci_server	P1000026
DatabaseServer1	Linux Red Hat	Enterprise	cmdb_ci_server	P1000199
DatabaseServer2	Linux Red Hat	Enterprise	cmdb_ci_server	P1000030
dbaix900nyc	AIX	5L	cmdb_ci_unix_server	P1000055
dbaix901nyc	AIX	5L	cmdb_ci_unix_server	P1000182
dbaix902nyc	AIX	5L	cmdb_ci_unix_server	P1000070
FileServerFloor1	Windows XP	Professional	cmdb_ci_server	P1000212
FileServerFloor2	Windows XP	Professional	cmdb_ci_server	P1000086
INSIGHT-NY-03	Windows XP		cmdb_ci_server	P1000201
lawson_app_100	Solaris	10	cmdb_ci_unix_server	P1000066
lawson_db_100	Solaris	10	cmdb_ci_unix_server	P1000176
Inux100	Linux Red Hat	2.6.9-22.0.1.ELsmp	cmdb_ci_linux_server	P1000165
Inux101	Linux Red Hat	2.6.9-22.0.1.ELsmp	cmdb_ci_linux_server	P1000054
MailServerUS	Solaris		cmdb_ci_server	P1000024
OWA-SD-01	Windows 2003 Standard		cmdb_ci_win_server	P1000116
PS LinuxApp01	Linux Red Hat	2.6.9-22.0.1.ELsmp	cmdb_ci_linux_server	P1000091
PS LinuxApp02	Linux Red Hat	2.6.9-22.0.1.ELsmp	cmdb_ci_linux_server	P1000207
PS LoadBal01	Windows 2003 Standard		cmdb_ci_win_server	P1000124
SAP AppSRV01	AIX		cmdb_ci_aix_server	P1000010
SAP AppSRV02	AIX		cmdb_ci_aix_server	P1000205
SAP LoadBal01	Windows 2003 Standard		cmdb_ci_win_server	P1000196

Select Server Class

cmdb_ci_aix_s...
cmdb_ci_linux_...
cmdb_ci_server

cmdb_ci_unix_...
cmdb_ci_win_s...

Data
Search

cmdb_ci_server
result.asset_tag
result.asset.link
result.asset.value
result.assigned
result.assigned_to
result.assignment_group
result.attestation_score
result.attestation_status
result.attested
result.attested_by
result.attested_date
result.attributes
result.business_unit
result.can_print
result.category
result.cd_rom
result.cd_speed
result.change_control
result.chassis_type
result.checked_in
result.checked_out
result.classification

15. Filter the data using slicer

16. Add the card to find the count of the each server class from ServiceNow

result.name	result.os	result.os_version	result.sys_class_name	result.asset_tag
ApplicationServerHelpdesk	Windows XP		cmdb_ci_server	P1000027
ApplicationServerPeopleSoft	Linux Red Hat		cmdb_ci_server	P1000204
AS400	OS/400	V5R2	cmdb_ci_server	P1000034
Car-1	Windows XP	5.2 build 3790	cmdb_ci_server	P1000171
Car-2	HP/UX	2.6.050	cmdb_ci_server	P1000071
Car-3	Windows XP	5.2 build 3790	cmdb_ci_server	P1000113
Car-4	Windows 2000	5.0 build 2195	cmdb_ci_server	P1000026
DatabaseServer1	Linux Red Hat	Enterprise	cmdb_ci_server	P1000199
DatabaseServer2	Linux Red Hat	Enterprise	cmdb_ci_server	P1000030
FileServerFloor1	Windows XP	Professional	cmdb_ci_server	P1000212
FileServerFloor2	Windows XP	Professional	cmdb_ci_server	P1000086
INSIGHT-NY-03	Windows XP		cmdb_ci_server	P1000201
MailServerUS	Solaris		cmdb_ci_server	P1000024
SD1	Windows XP		cmdb_ci_server	P1000198
Service-now Production Sacramento	Linux Red Hat	Enterprise Server 3	cmdb_ci_server	P1000173
Service-now Production San Diego	Linux Red Hat	Enterprise Server 3	cmdb_ci_server	P1000114
VMWARE-SD-04	Windows 2000 Server		cmdb_ci_server	P1000095
VMWARE-SD-07	Windows 2000 Server		cmdb_ci_server	P1000021
Webserver FLX			cmdb_ci_server	P1000242

Select Server Class

cmdb_ci_aix_s...

cmdb_ci_linux_...

cmdb_ci_server

cmdb_ci_unix_...

cmdb_ci_win_s...

result.name	result.os	result.os_version	result.sys_class_name	result.asset_tag
ApplicationServerHelpdesk	Windows XP		cmdb_ci_server	P1000027
ApplicationServerPeopleSoft	Linux Red Hat		cmdb_ci_server	P1000204
AS400	OS/400	V5R2	cmdb_ci_server	P1000034
Car-1	Windows XP	5.2 build 3790	cmdb_ci_server	P1000171
Car-2	HP/UX	2.6.050	cmdb_ci_server	P1000071
Car-3	Windows XP	5.2 build 3790	cmdb_ci_server	P1000113
Car-4	Windows 2000	5.0 build 2195	cmdb_ci_server	P1000026
DatabaseServer1	Linux Red Hat	Enterprise	cmdb_ci_server	P1000199
DatabaseServer2	Linux Red Hat	Enterprise	cmdb_ci_server	P1000030
FileServerFloor1	Windows XP	Professional	cmdb_ci_server	P1000212
FileServerFloor2	Windows XP	Professional	cmdb_ci_server	P1000086
INSIGHT-NY-03	Windows XP		cmdb_ci_server	P1000201
MailServerUS	Solaris		cmdb_ci_server	P1000024
SD1	Windows XP		cmdb_ci_server	P1000198
Service-now Production Sacramento	Linux Red Hat	Enterprise Server 3	cmdb_ci_server	P1000173
Service-now Production San Diego	Linux Red Hat	Enterprise Server 3	cmdb_ci_server	P1000114
VMWARE-SD-04	Windows 2000 Server		cmdb_ci_server	P1000095
VMWARE-SD-07	Windows 2000 Server		cmdb_ci_server	P1000021
Webserver FLX			cmdb_ci_server	P1000242

Select Server Class

cmdb_ci_aix_s...

cmdb_ci_linux_...

cmdb_ci_server

cmdb_ci_unix_...

cmdb_ci_win_s...

Server Count

19

References:

- https://www.servicenow.com/docs/bundle/yokohama-api-reference/page/integrate/inbound-rest/concept/c_RESTAPI.html
- <https://developer.servicenow.com/dev.do#!/reference/api/yokohama/rest/>
- <https://www.servicenow.com/docs/bundle/washingtondc-api-reference/page/build/applications/concept/api-rest.html>
- <https://learn.microsoft.com/en-us/power-bi/>
- <https://learn.microsoft.com/hi-in/power-bi/fundamentals/>

Thank you

Vijay Yacham